HP at AraGenre 2026: LLM-Guided Pseudo-Labeling for Hierarchical Arabic Genre Classification with MARBERT

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Abstract

This paper presents the system submitted by team HP for the AraGenre 2026 Shared Task on Hierarchical Definition-Guided Arabic Genre Classification. To address the low-resource constraint of the provided dataset (140 labeled training samples), we propose a two-stage pipeline combining LLM-based pseudo-labeling with supervised fine-tuning. In the first stage, we evaluate several models under a 5-fold cross-validation framework, including Gemini 3.5 Flash-Lite (0.9413), Qwen2.5 7B (0.7203), LLaMA 3.1 8B (0.6380), and a multilingual NLI model (0.3099). Based on these results, Gemini 3.5 Flash-Lite is selected to generate pseudo-labels for the unlabeled test set using a definition-guided few-shot prompt. In the second stage, MARBERTv2 and ARBERTv2 are fine-tuned on approximately 28,000 pseudo-labeled samples. Our experiments show that MARBERTv2 achieves the best standalone score of 0.7169 Hierarchical Macro F1, outperforming both ARBERTv2 (0.7118) and a soft voting ensemble (0.7165), suggesting that Arabic-specialized pretraining of MARBERTv2 is more effective than model ensembling for this task.

1 Introduction

The AraGenre 2026 Shared Task challenges participants to perform hierarchical genre classification on Arabic texts, identifying both broad and fine-grained specific communicative functions (El-Haj et al., 2026). A primary challenge of this task is its extreme low-resource setting: the provided training data consists of only 140 samples, requiring systems to generalize across a large taxonomy of 74 specific genres on a hidden evaluation benchmark of approximately 28,000 sentences.

Standard supervised learning approaches typically struggle to avoid overfitting under such severe data scarcity. To bridge this gap, we adopt a pseudo-labeling approach (Lee, 2013). We leverage a Large Language Model (LLM) as an annotator to label the large unannotated test set, followed by fine-tuning dedicated Arabic encoders on this augmented corpus. Through a rigorous 5-fold cross-validation framework on the training data, we identified Gemini 3.5 Flash-Lite as our optimal annotator, achieving the highest Hierarchical Macro F1 score (0.9413) while offering the scalability required for large-scale inference.

Our main contributions are as follows:

- We evaluate various LLMs and NLI models in a low-resource Arabic genre classification setting, demonstrating the superiority of Gemini 3.5 Flash-Lite in few-shot definition-guided reasoning.
- We design a scalable pseudo-labeling pipeline to automatically annotate approximately 28,000 test samples.
- We demonstrate that fine-tuning MARBERTv2 on these pseudo-labels achieves a Hierarchical Macro F1 of 0.7169 on the blind test set, outperforming both ARBERTv2 and a soft-voting ensemble. Our implementation is publicly available[1].

2 Background

2.1 Task Setup and Dataset

The AraGenre 2026 Shared Task focuses on hierarchical Arabic genre classification (El-Haj et al., 2026). Given an Arabic text segment, ranging from short fragments to longer passages and encompassing Modern Standard Arabic, Classical Arabic, and various dialects, systems must predict two levels of communicative function: a Broad Genre and a fine-grained Specific Genre (El-Haj et al., 2026). For example, an input text such as “ ...” (Recent

[1] https://github.com/phonghoangKHDL/AraGenre-HP-2026

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economic data indicates...) must be classified into the Broad Genre of *Informative* and the Specific Genre of *Analytical Reports*.

The provided dataset simulates extreme low-resource conditions, with the training set containing only 140 labeled instances (El-Haj et al., 2026). The official taxonomy defines 6 broad categories (Informative, Interactive, Learning, Legal, Creative, Religious), further subdivided into 74 specific genres. The hidden evaluation benchmark comprises approximately 28,000 naturally occurring Arabic texts with noisy content and diverse writing styles (El-Haj et al., 2026).

2.2 Related Work

Arabic Language Models. Transformer-based encoder models have significantly advanced Arabic Natural Language Processing. MARBERTv2 (Elmadany et al., 2022) is pretrained on large-scale dialectal Arabic data, while ARBERTv2 (Elmadany et al., 2022) is trained on 243 GB of Modern Standard Arabic text. Both models have demonstrated robust performance across a wide range of Arabic classification tasks.

LLM-Guided Data Augmentation. In low-resource scenarios, recent studies have increasingly adopted Large Language Models as zero-shot or few-shot annotators to automatically generate synthetic data or pseudo-labels (Lee, 2013). By leveraging the broad semantic knowledge of LLMs to supervise smaller, task-specific encoders, researchers can achieve competitive performance without the inference costs associated with deploying large generative models at test time. Our work follows this paradigm by using Gemini 3.5 Flash-Lite (Google, 2026a) as a pseudo-labeler to fine-tune MARBERTv2 (Elmadany et al., 2022) and ARBERTv2 (Elmadany et al., 2022), targeting the specific challenge of hierarchical Arabic genre classification.

3 System Overview

Our methodology addresses the severe data scarcity by framing the problem as a pseudo-labeling pipeline. We utilize a powerful LLM to generate pseudo-labels for the unannotated test set, which are subsequently used to supervise dedicated Arabic encoders.

3.1 Prompt Engineering and Model Selection

To select the optimal LLM annotator, we evaluated several models, including LLaMA 3.1 8B (Grattafiori et al., 2024), Qwen2.5 7B (Team, 2024), and Gemini 3.5 Flash-Lite (Google, 2026a) [2]. Although a development set of 110 sentences is provided, its gold labels are not publicly released, precluding offline evaluation. We therefore conducted 5-fold cross-validation directly on the 140-sample training set.

A primary challenge when prompting LLMs for hierarchical classification is label hallucination — where the model generates plausible but out-of-taxonomy categories. To mitigate this, we designed a strict definition-guided prompt. The system instruction embedded the complete hierarchical taxonomy: 6 broad genres and 74 specific genres, alongside their official English definitions. Furthermore, we dynamically constructed balanced few-shot examples (Brown et al., 2020) for each fold to provide in-context learning without causing data leakage. Under this prompt constraint, Gemini 3.5 Flash-Lite successfully avoided label hallucination and achieved a Hierarchical Macro F1 score of 0.9413 in cross-validation.

3.2 Large-Scale Pseudo-Labeling

When scaling the pseudo-labeling process to the hidden evaluation benchmark (27,972 sentences), practical engineering constraints become equally important as empirical performance. Standard generative models, including Gemini Flash and Pro variants or other commercial APIs, are typically capped at 10,000 or fewer requests per day, making single-run inference infeasible for a dataset of this scale. Gemini 3.5 Flash-Lite addresses this bottleneck by supporting up to 150,000 requests per day under the free Tier 1 quota (Google, 2026b). Furthermore, it is highly optimized for cost-efficiency compared to heavier commercial counterparts. Combined with its strong cross-validation accuracy, it represents the optimal choice for large-scale pseudo-labeling.

To optimize the API interaction, we leveraged context caching. By caching the system instruction containing the 74 definitions and few-shot examples, we significantly reduced inference latency and token costs across the 28,000 requests. During post-processing, we applied a strict hierarchical

[2] API model identifier: `gemini-3.5-flash-lite`, accessed via the Google Generative AI Python SDK (`google-generativeai 0.8.6`).

constraint: if a predicted specific genre did not align with its parent broad genre, the broad genre was deterministically corrected based on the official taxonomy mapping.

3.3 Fine-Tuning on Pseudo-Labeled Data

It is worth noting that our pipeline follows a transductive learning paradigm: Gemini 3.5 Flash-Lite labels the hidden test inputs directly, and the fine-tuned encoders are subsequently trained and evaluated on those same inputs. To empirically validate this design choice, we first submitted Gemini 3.5 Flash-Lite's direct predictions as a baseline, which achieved a Hierarchical Macro F1 of 0.5724 on the official benchmark. This result motivated the pseudo-labeling stage: by using these predictions as training signal for Arabic-specialized encoders, we achieved 0.7169 — an absolute improvement of 0.1445. We attribute this gain to Arabic-specialized pretraining, which provides a stronger prior for the noisy, naturally occurring test distribution.

We filtered the pseudo-labeled test set, discarding a minimal number of structural failures or "Unknown" outputs, resulting in a training corpus of 27,888 pseudo-labeled samples spanning all 74 specific genres. We fine-tuned two Arabic-specialized language models: MARBERTv2 (Elmadany et al., 2022) and ARBERTv2 (Elmadany et al., 2022). The models were trained to predict the 74 specific genres directly, with the broad genres inferred deterministically during inference. To accelerate training on this large corpus, we implemented dynamic padding via a data collator and restricted the maximum sequence length to 256 tokens.

3.4 Ensemble Strategy

We implemented a soft voting ensemble to explore system robustness. We extracted the raw logit matrices (dimensions: $27{,}972 \times 74$) from both the fine-tuned MARBERTv2 and ARBERTv2 models. These logits were converted into probability distributions using the Softmax function and averaged. The final predictions were derived from the argmax of this combined distribution.

4 Experimental Setup

4.1 Data Utilization

We utilized the official AraGenre 2026 dataset (El-Haj et al., 2026). The provided 140 labeled training samples were used exclusively for few-shot prompt construction (Brown et al., 2020) and baseline cross-validation. Although a development set of 110 samples was provided, the lack of publicly available gold labels prevented its use for offline evaluation. The hidden evaluation benchmark, comprising 27,972 unannotated texts, was used for pseudo-labeling and final testing.

Table 1: Hyperparameter settings for encoder fine-tuning.

Hyperparameter	**Value**
Epochs	3
Learning Rate	2×10^{-5}
Weight Decay	0.01
Per-Device Batch Size	8
Gradient Accumulation Steps	2
Effective Batch Size	16
Maximum Sequence Length	256
Precision	Mixed (FP16)
Optimizer	AdamW
Loss Function	Cross-Entropy

4.2 Implementation Details

For the generative baselines, API inference with Gemini 3.5 Flash-Lite was conducted with a temperature of $T = 0.0$ for deterministic outputs. For encoder fine-tuning, we utilized the Hugging Face `transformers` and `datasets` libraries. MARBERTv2 and ARBERTv2 were fine-tuned for 3 epochs—a standard practice for BERT-based architectures to prevent overfitting on pseudo-labeled data—with a learning rate of 2×10^{-5}, weight decay of 0.01, and an effective batch size of 16 (achieved using a per-device batch size of 8 with 2 gradient accumulation steps). We applied mixed precision (FP16) and dynamic padding via a data collator, restricting the maximum sequence length to 256 tokens to optimize memory usage and accelerate training. Table 1 details the training configurations used for fine-tuning the Arabic encoder models (MARBERTv2 and ARBERTv2) on the pseudo-labeled dataset.

4.3 Evaluation Metrics and Objective

Systems are evaluated at both the broad and specific genre levels using Macro F1, which treats all classes equally regardless of support size. The official ranking metric for the shared task is the Hierarchical Macro F1, which is calculated as the unweighted average of the Broad Macro F1 and the

Table 2: The performance on 5-Fold cross-validation results on the training set.

Model	Broad F1	Specific F1	Hierarchical F1
mDeBERTa-v3 (NLI)	0.3406	0.2792	0.3099
LLaMA 3.1 8B	0.6431	0.6329	0.6380
Qwen2.5 7B	0.7431	0.6975	0.7203
Gemini 3.5 Flash-Lite	**0.9413**	**0.9414**	**0.9413**

Table 3: Final system performance on the hidden evaluation benchmark.

Model	Broad F1	Specific F1	Hierarchical F1
Gemini 3.5 Flash-Lite	0.7596	0.3853	0.5724
ARBERTv2	0.8933	0.5302	0.7118
Soft Voting Ensemble	0.8970	0.5359	0.7165
MARBERTv2	**0.9000**	**0.5339**	**0.7169**

Specific Macro F1:

$$F1_{Hierarchical} = \frac{F1_{Broad} + F1_{Specific}}{2}$$

We formulate this as a standard multi-class classification problem over 74 discrete categorical classes. Our MARBERTv2 and ARBERTv2 encoders are optimized using the Cross-Entropy loss objective over the 74 specific genres. During inference, the specific genre is predicted via argmax, and its corresponding broad genre is deterministically inferred based on the official taxonomy mapping, ensuring structural consistency with the hierarchical evaluation metric.

5 Results and Analysis

5.1 Baseline Evaluation

Table 2 presents the 5-fold cross-validation performance of our baseline models on the 140-sample training set. Gemini 3.5 Flash-Lite significantly outperformed the other models, achieving a Hierarchical Macro F1 of 0.9413. The zero-shot NLI approach using mDeBERTa-v3 (He et al., 2023) struggled to capture the semantic nuances of the 74 specific genres, yielding only 0.3099.

5.2 Final System Performance

Table 3 presents the performance of our models on the official hidden evaluation benchmark. Our fine-tuned MARBERTv2 standalone model achieved our highest performance with a Hierarchical Macro F1 of 0.7169. By comparison, ARBERTv2 achieved 0.7118, and the soft voting ensemble yielded 0.7165. The slight performance drop in the ensemble suggests that MARBERTv2's dialectal pretraining provides a stronger signal for this task than model combination.

5.3 Error Analysis

An analysis of the cross-validation results for the open-source baselines reveals a consistent pattern of confusion among semantically overlapping genres. Both LLaMA 3.1 8B and Qwen2.5 7B struggled significantly with the *Community Discussions* genre, recording a recall of only 0.30 despite perfect precision, suggesting that the models rarely predicted this genre and systematically misclassified its instances elsewhere. This is further evidenced by the high recall but low precision observed for *Personal Narratives* (LLaMA: 0.71, Qwen: 0.56). For instance, a training sample (ID: `train_000069`) belonging to *Community Discussions*, translated as *"From my personal experience, the matter just requires patience at the beginning."* was consistently misclassified as *Personal Narratives* by both open-source baselines, likely due to its explicit first-person framing and subjective tone, surface features shared by both genres. Similarly, *Official Statements* achieved low recall in both models (LLaMA: 0.25, Qwen: 0.40), suggesting difficulty in distinguishing formal institutional language from adjacent genres such as *Administrative Regulations*. These patterns highlight the inherent difficulty of fine-grained genre classification when communicative intents overlap in informal or formal discourse.

6 Conclusion

We addressed the extreme low-resource challenge of the AraGenre 2026 shared task by combining LLM-guided pseudo-labeling with encoder fine-tuning. By leveraging Gemini 3.5 Flash-Lite's generous daily request quota and definition-guided reasoning capabilities, we successfully annotated approximately 28,000 test samples to fine-tune MARBERTv2 and ARBERTv2. This approach proved effective, achieving a Hierarchical Macro F1 of 0.7169. Future work will explore integrating retrieval-augmented generation (RAG) to explicitly align external genre definitions during the encoder fine-tuning phase.

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A Prompt Design for Pseudo-Labeling

To guide the Gemini 3.5 Flash-Lite model in generating accurate pseudo-labels without hallucinating out-of-taxonomy classes, we utilized a strict definition-guided prompt. The prompt enforced a JSON output structure and was executed with a temperature of 0.0 to ensure deterministic generation. The system instruction was structured as follows:

> **System Instruction:**
> You are an expert Arabic genre classifier.
>
> INSTRUCTIONS:
>
> 1. Read the Arabic text carefully.
> 2. Identify the broad genre based on communicative function.
> 3. Select the specific genre strictly from its subcategories.
> 4. Output ONLY valid JSON: `{"broad_genre": "...", "specific_genre": "..."}`
>
> **Hierarchical Taxonomy & Definitions**
> [*List of 6 Broad Genres and 74 Specific Genres with their official English definitions*]
>
> **Balanced Few-shot Examples**
> [*Dynamically sampled examples mapping Arabic text to its Broad and Specific Genre*]

For the hidden evaluation benchmark, this entire prompt payload was cached using the Gemini Context Caching API with a Time-To-Live (TTL) of 28,800 seconds to reduce token costs across the approximately 28,000 inference requests.